POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Problem Solving Technique | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A certain number k is a multiple of 9. Add the digits together. If the result has more than one digit, add those together. Continue adding digits together until you have a one digit answer. It will be a 9. Explain why this is so? 8

OR

A watermelon weighs 500 pounds. It turns out that 99% of the weight of the watermelon is due to water in the watermelon. After the watermelon has sat in a drying room for a while, it turns out that it is only 98% water by weight. How much does it weigh now?

b) A ten foot pole is dropped into a milling saw and randomly cut into three shorter poles. What is the probability that these three pieces will form a triangle? 7

2. a) A right-angled triangle has sides of length l, m 10. Note that 10 is not the hypotenuse, and the both l and m are integers. Given this information, find l, m. 8

b) A game is played by two players. They begin with a pile of thirty chips, all the same. For his or her move, a player may remove 1 to 6 chips. The player who removes the last chip wins. What strategy can the first player use so that he will always win? 7

3. a) Solve the crypto arithmetic problem: 7

NUDE+NOT+RUDE+NOR=CRUDE

b) Six people, named A, B, C, D, E, F are in the dining car of a train. They are one each from New York City, Chicago, Tulsa, St. Louis, Milwaukee and Atlanta. The following facts are known. 8

i. A and the man from New York City are physicians.

ii. E and the woman from Chicago are teachers.

iii. The person from Tulsa and C are engineers.

iv. B and F are veterans of Gulf war, but the person from Tulsa has never served in the military.

v. The person from Milwaukee is older than A.

vi. The person from ~~Milwaukee~~ Atlanta is older than ~~A~~ C

vii. At St. Louis, B and the man from New York get off.

viii. At San Francisco, C and the man from Milwaukee gets off. Match the names of the people with their professions and their cities.

4. a) Consider a unit cube with four of its eight vertices joined to form a regular tetrahedron with vertices A,B,C and D. What is the ratio of the surface area of the cube to the surface area of the tetrahedron? 7

b) A 10 years old child puts Rs. 100000 in the bank. She intended with withdraw the money on her 21st birthday. Which one scheme is better for her? 8

i. An account with 5% interest compounded daily

ii. An account with 5.1% interest compounded weekly.

5. a) Suppose that you have 9 pearls. They all look the same, but 8 of have equal weight and one is different. The odd pearl is either heavier or lighter; you do not know which. The only equipment that you have at hand is a balance scale. How can you use the scale to find the odd pearl in just three weighing? 7

b) It begins snowing some time before noon. The snow fall steadily, when measured by the rate of change of depth. At exactly noon, a snow plow begins working at a steady rate (in terms of cubic feet of snow removed per hour). The plow clears two blocks during the first hour of work, and one block during the second hour. At what time did it begin snowing? 8

OR

Examine the equations:

$$1 = 1$$
$$2+3+4 = 1+8$$
$$5+6+7+8+9 = 8+27$$
$$10+11+12+\ldots+16 = 27+64$$

Determine the pattern and prove the identity.

6. a) There are three married couples that need to cross a river. A small boat is available that will hold just two people at a time. The males involved are quite jealous. No woman can be left with a man unless her husband is also present. There are no other constraints. How can these six people cross the river? What is the fewest numbers of trips possible? 7

b) Imagine the election involving three candidates A,B,C. There are 33 Votes. Every voters vote and every voter gives a near ranking to the three candidates. The result is shown below. 8

Ranking	Number of Votes
ABC	10
ACB	4
BAC	2
BCA	7
CAB	3
CBA	7

We have three methods namely Plularity method (who get most votes is Winner) .Borda method (Candidates with best average is winner) and Hare Method (fewest vote candidates is eliminated and votes are shared to second rank candidate method).Show that which method favors which of the candidates.

7. Write short notes on: (**Any two**) 2×5

a) Explain impossible problems.

b) Determine how many zeros end the number 780! – 310!.

c) Is $10^{\frac{1}{10}} > 2^{\frac{1}{3}}$?